

Final Design Report: The ClampMaster 2000



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Executive Summary

In this report, we will be discussing a solution to help our user who has cerebral palsy that makes it difficult for her to get from place to place on her own. She is a 17-year-old coming to Northwestern in the Fall of 2024. While she is perfectly capable of walking from class to class across Northwestern's campus in theory, in practice, it would put too much strain on her and undo much of the physical therapy work that she has had. Thus, our job was to help facilitate a way for the user to create a low-profile, aesthetically pleasing, and easy-to-use solution that allows an individual with cerebral palsy to mount a bike safely, comfortably, and independently.

Abigail Jones, our user's physical therapist, wants a safe and reliable product to help her client ride a bike. Though she will not be using the product herself, she wants to make sure that her client can use it independently. This project has been brought to us by Shirley Ryan Ability Lab. They may want to reproduce our product for other patients who have similar conditions to our primary user.

Through our research, we have discovered existing solutions that would require our client to invest in a mobility scooter or bike, which is cumbersome, expensive, and not aesthetically pleasing. Further research showed that there are ways to get onto a bike with our patient's limitations, such as mounting a bike on an EV charger or bike lock station to hold it upright. However, the solutions go against our goals and hinder the client's independence.

The primary goal was to make it as easy as possible for her to get on and off her bike. After talking with the user and her therapist, we concluded that designing for a Divvy bike was the easiest way to test our potential design. We set off on our project, creating mockups and running user testing.

Our solution, the ClampMaster 2000, is a mechanism that inhibits the front wheel from rotating, allowing the user to put all of her weight on the handlebars without the contraption giving way or spinning unexpectedly. The ClampMaster 2000 works by having one clamp attached to the lower frame of the bike right next to the front wheel and a swinging arm. On the other end of the arm is another clamp used to clamp onto the front wheel. This allows the user to easily immobilize the front wheel and makes it easier for her to get on and off the bike. Additionally, this fulfills our requirements of sustainability, ease of use with one hand, portability, and being low-profile.

The ClampMaster 2000 successfully addresses the challenge of supporting a 17-year-old student with cerebral palsy in mounting a bike independently. With its three-component design, the device prevents handlebar rotation and ensures stability. Our team is excited about the potential future developments and the positive impact this solution can have on our client and others with similar needs.

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Introduction

In our research of ability bikes and mobility option for people with disabilities, we found that prohibitive cost, overspecified designs, and limited supply has made it so many people with mobility issues, ranging from arthritis to cerebral palsy to injury, are unable to access to many low speed man-powered vehicles that individuals without these limitations take for granted. Things like bikes, scooters, and skateboards are almost impossible to use for people with mobility problems, which inspired our further research and product development of our solution for our specific client.

Our user is a 17 year old coming to Northwestern in the Fall of 2024. The user has cerebral palsy among other motor issues that makes it difficult for her to get from place to place on her own. While she is perfectly capable of walking from class to class across Northwestern's Campus in theory, in practice, it would put too much strain on her and undo much of the physical therapy work that she has had and would end up hurting her. Thus, our job was to help facilitate a way for the user to create a low-profile, aesthetically pleasing, and easy-to-use solution that allows an individual with cerebral palsy to mount a bike safely, comfortably, and independently.

Through our secondary research and client interviews, we discovered that a solution exists that would require our client to invest in a mobility scooter or bike, which is cumbersome, expensive, and not aesthetically pleasing. Further research showed that there are ways to get onto a bike with our patient's limitations, such as mounting a bike on an EV charger or bike lock station to hold it upright. However, the former solution goes against our every goal from our mission statement, and the latter hinders the clients independence by attaching her to specific points she has to reach to get on and off the bike, limiting where she can go and making many trips more inconvenient than they otherwise would have to be.

The primary goal was to make it as easy as possible for her to get on and off of her bike. After talking with the user and her therapist, we concluded that designing for a divvy bike was the easiest way to be able to test our potential design. Furthermore, we set off on our project, creating mockups and running user testing detailed further in Appendix H.

Our solution–The ClampMaster 2000– is a mechanism that inhibits the front wheel from rotating, allowing the user to put all of her weight on the handlebars without the contraption giving way or spinning unexpectedly. The ClampMaster 2000 works by having one clamp attached to the lower frame of the bike right next to the front wheel and a swinging arm that has another clamp that the user can clamp onto the front wheel. This allows the user to easily immobilize the front wheel and makes it easier for her to get on and off the bike. Additionally, this fulfills our requirements of sustainability, ease of use with one hand, portability, and being low-profile.

This report documents our user and their specific needs and requirements, transitions into how we designed for those specific requirements and why we have the features we chose, and future developments looking beyond the scope of the class.

Users and Requirements

Primary User

Our main user, a 17-year-old currently studying in high school, is coming to Northwestern next year to study Chemistry. She has cerebral palsy and is determined to ride a bike to get to and from class. She is 74 kg, 4'9", 4'11" with her AFOs (ankle-foot orthotics) that stabilize her. Currently, she is working with a physical therapist to extend her range of motion to safely mount a bike. She likes darker colors and wants her bike to be a single solid color. She prefers that our design does not stand out when it is on the bike.

Secondary User

Abigail Jones, our user's physical therapist, wants a safe and reliable product to help her client ride a bike. Though she will not be using the product herself, she wants to make sure that her client can use it independently.

Tertiary User

This project has been brought to us by Shirley Ryan Ability Lab. They may want to reproduce our product for other patients that have similar conditions to our primary user.

Requirement 1: Stability

The user will be holding herself up using the handlebars therefore the product must be able to stabilize the handlebars and keep them from rotating. It must safely carry her body weight since she will only be supporting herself using the handlebars. In the case that the product fails or yields, the user may lose her balance and injure herself therefore it must be reliable and the tolerances must be small.

Requirement 2: Easy, One Handed Use

The user is going to be using her bike every day to get to and from class therefore the product has to be easy to use. No tools should be needed on a daily basis to use the product. Anything that has to be tightened or clamped must have large knobs or handles to allow the user to get the necessary leverage. She will also be holding on to the handlebars with one hand at all times for stability therefore the product must be easily usable with one hand. Especially when she is on the bike and wants to get off, she may need to reach for our product while still sitting on the bike. To ensure that she doesn't fall off, she must hold on to the bike with one hand as she uses our product.

Requirement 3: Durability

Since the product will remain outside on a bike and will be used every day, it must be weatherproof, corrosion-resistant, and durable. The product must not require constant maintenance and must last at least 4 years.

Requirement 4: Adjustability

Since the user does not currently have a bike and has not decided on a model yet, the product must be adjustable to any bike that she chooses to buy in the future. This also allows for the product to be used by other people with similar conditions who have different bikes.

Requirement 5: Portability

The product needs to be easily portable, allowing for the user to easily remove and replace the product if necessary. Additionally, this creates constraints upon the size and weight of the product.

Requirement 6: Noticeability and Aesthetics

The primary user wants a low-profile product that does not attract attention when in use. Therefore the product must be small in size and must blend in with the rest of the bike. She also wants the bike to look nice as a whole and likes solid dark colors. The final product must be visually appealing.

Design Concept and Rationale

The concept of the ClampMaster 2000 began as a collection of our various initial mockups. These mockups included things like restraining the swivel by connecting to the front wheel, using friction with the floor, or a set of bars sticking into the frame, or a wooden bar dropping to the floor. However, after further iteration and testing with the user, we discovered that many such mockups failed to meet our initial requirements for stability and ease-of-use. Therefore, our team came together to implement the promising parts of our initial mockups in order to create a more stable and easy-to-use solution, the ClampMaster 2000. Consisting of three main parts, the ClampMaster2000 is a clamp that attaches to the frame of the bike and clamps onto the front wheel which ultimately stops it from rotating. This is to help our client use the front of the bike as a stabilizing handle in order to mount the bike safely. This fulfills the requirements of being secure, easy to use, and a subtle way to stop the front wheel from rotating, while also showcasing some features, like the ability to be used with only one hand, being able to stay on the bike and not have to be taken off, and the ability to stay out of the way of the users legs while pedaling by clamping onto the frame.

The three main components of the ClampMaster 2000 are: a grip that goes around the frame, the clamp itself, and a rotating axle connecting the two parts. A sketch of the prototype is shown in Figure 1. These parts are designed in such a way that it attracts very little attention and keeps the front of the bike steady.

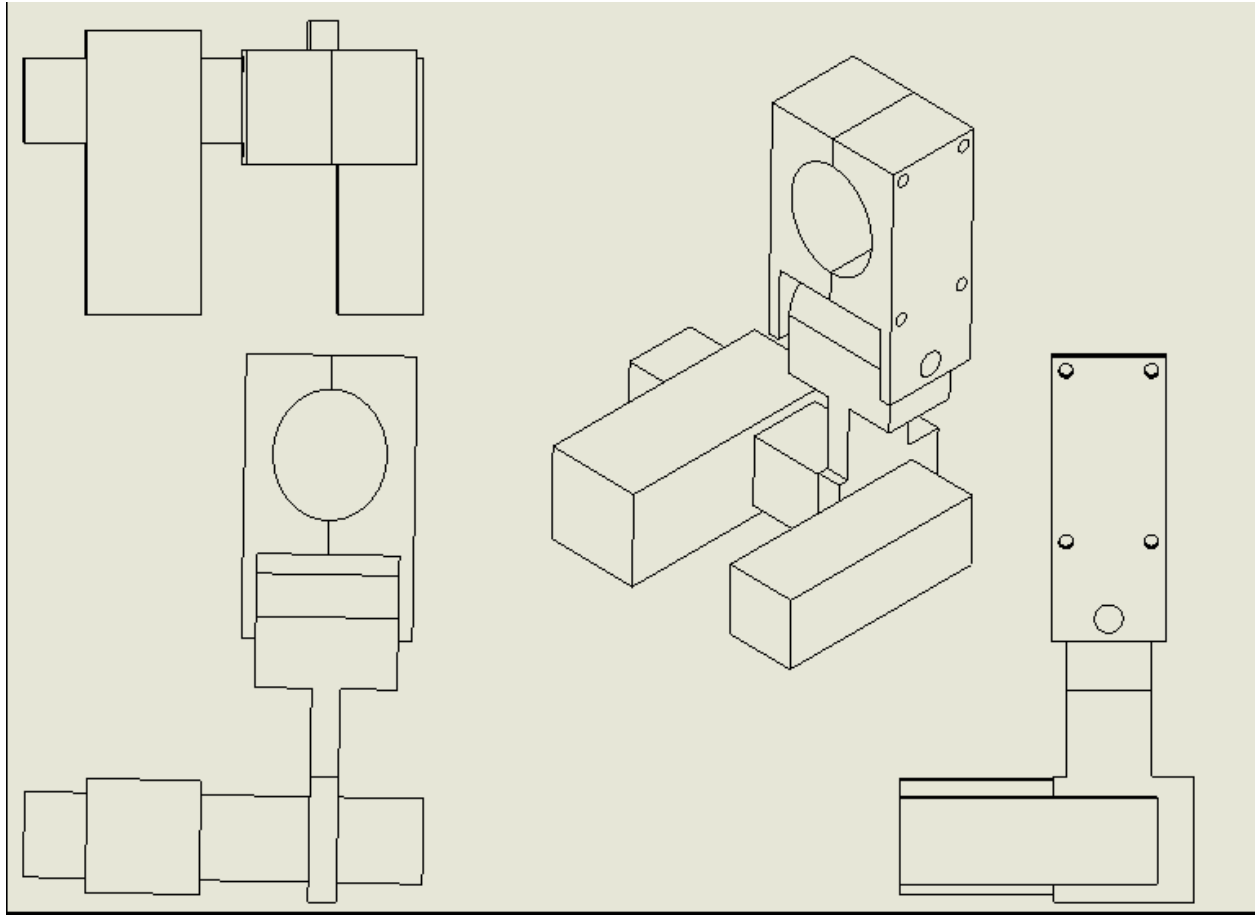


Figure 1: This figure shows the isometric and orthographic sketches of the prototype of the ClampMaster 2000

Connector

The first main component of the design is the connector, the part that goes around the frame. This secures the entire design onto the bike by placing this around the frame of the bike. This is actually two symmetrical parts, each part 2 inches wide, and once placed around the frame, the two parts are screwed together with screws with a diameter a half inch wide, once a quarter of an inch from the top of the connector and once near the bottom, a quarter of an inch from where the rotatable axis is connected to the connector. Once screwed on, it will stay in place on the frame. This is because an oval is formed when combining the two symmetrical parts together, with a height of 2.3 inches and a width of 2 inches. This is so the connector could fit around the frame of a Divvy bike which will be used to showcase the product. At the time at which our team had met with our client, she had done most of her practice of getting onto the bike with a Divvy bike.

Due to this, our team decided to test our prototype on a Divvy bike. The frame for a Divvy bike is bigger than the frame of the tricycle our client is planning on getting. Therefore, in order to fit the connector onto a Divvy, the oval had to be big enough to fit onto the frame of a Divvy. While this is fitted for a Divvy bike, it can still fit around a normal bike or tricycle frame. However, it wouldn't fit as well on a normal bike or tricycle frame as it would on a Divvy bike.

Underneath the oval, on each side there is a small rectangular extension. That extension is where the rotatable axis goes, connected onto it with a screw of half an inch of diameter .4 inches from the bottom. This was placed here in order to ensure that the rotatable axis doesn't get in the way of the connector.

While our client said that she was fine doing some work, she would like it if the amount of work she needed to do was reduced as much as possible. Therefore, our team reasoned that a good feature of the ClampMaster 2000 would be that she only had to attach the device onto the bike once, and after that the device would be able to stay on the bike. That is why the device consists of two pieces that can be screwed around the frame of the bike, and once placed, stay on the bike.

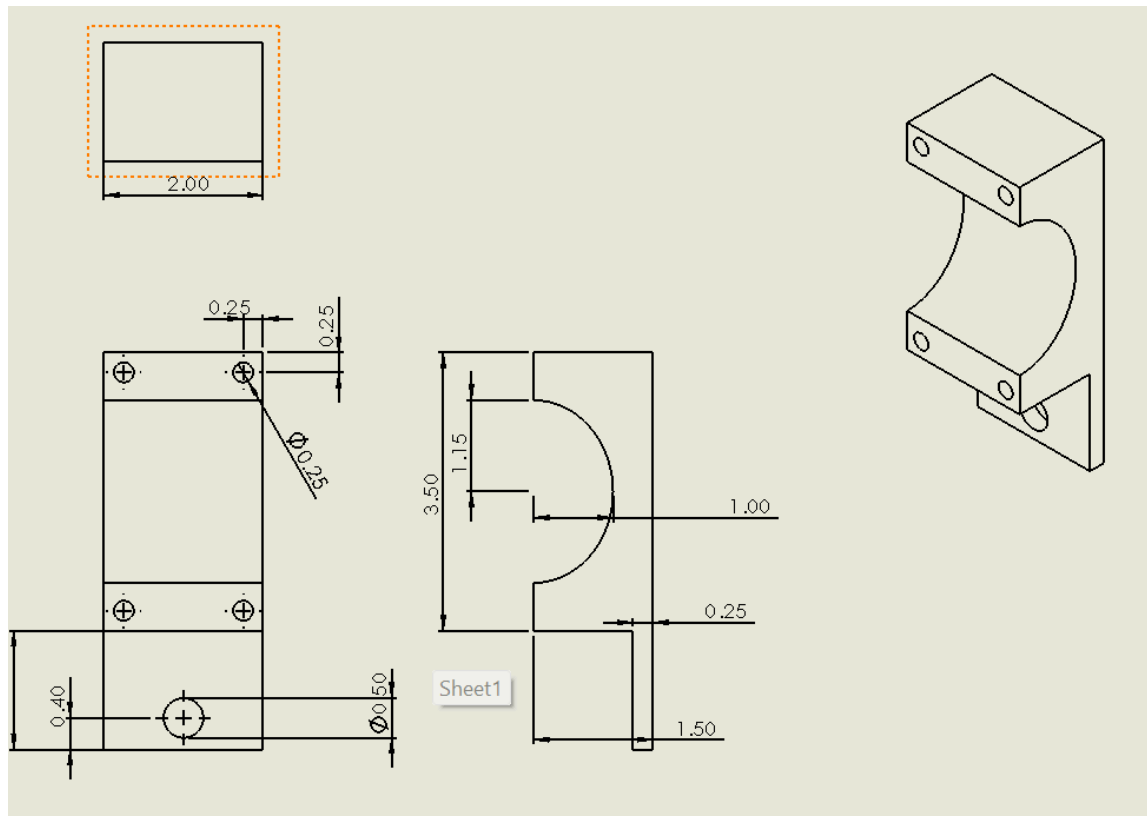


Figure 2: Measurements of ClampMaster 2000 Connector

Rotating Axle

The second main component is the rotating axle, which connects to the connector and the clamp. This axle serves the purpose of moving the clamp on and off of the wheel. With a width of half an inch at the thinnest section, and a length from the screw where it was connected to the end of the axle of 4.5 inches (as shown in Figure 2), this axle is very adjustable and capable of rotating 180 degrees.

These dimensions allow for a nice feature of the ClampMaster 2000, which is that the rotatable axis can swing backwards onto the frame and clamp onto it, keeping the device out of the way of the client's feet while pedaling (see more explanation in the "Clamping Device" section).

During user testing, we witnessed how far the client is able to bend down. While the client would be able to bend down far enough to just slide the clamp out of the way, the rotating axle decreases the amount that she would have to bend down when using the clamp. Instead of bending down and having to slide it back, possibly unstabilizing her, she can just bend down, grab the clamp and then move it back up the frame, where it can be easily clamped onto the frame. This rotatable axis allows for that by very easily being movable and not requiring much force to rotate

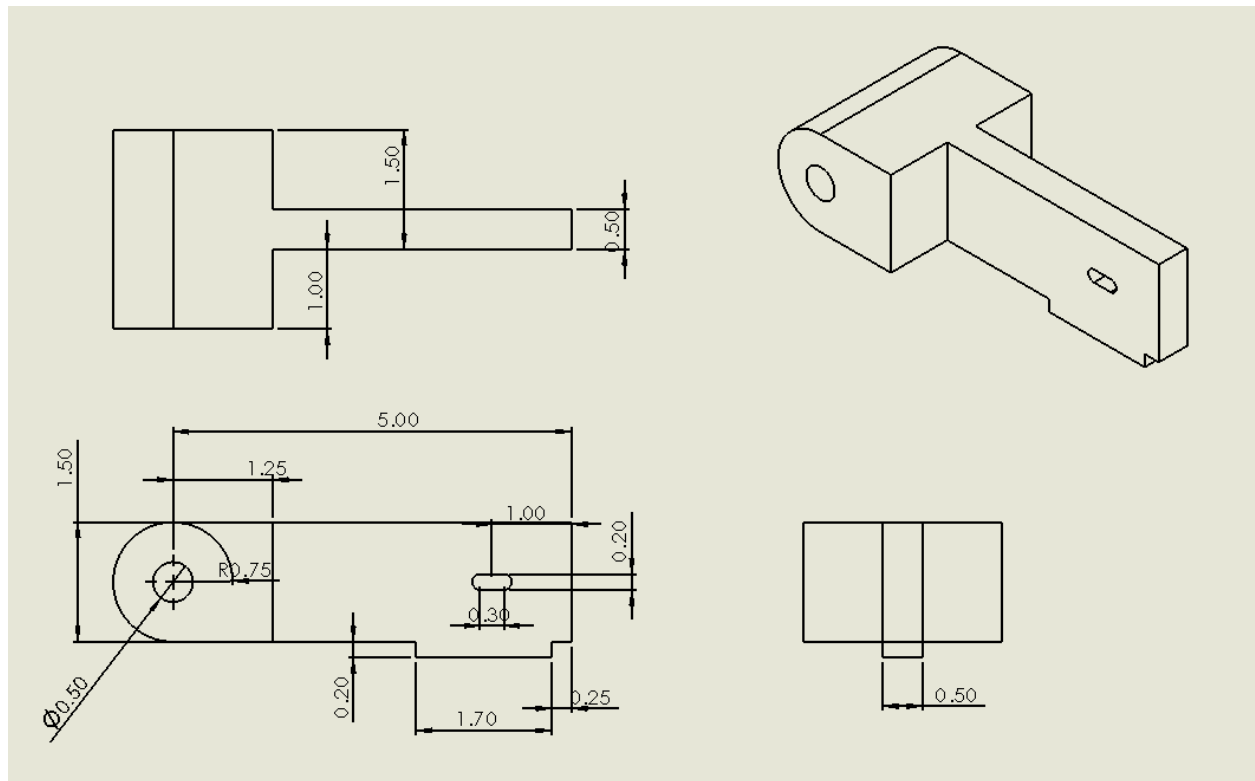


Figure 3: Measurements of ClampMaster 2000 Rotating Axle

Clamping Device

The third main component of the ClampMaster2000 is the clamp, which is an Irwin Quick Grip clamp that can clamp onto the front wheel of the bike and keep it stationary. This kind of clamp was chosen for the sake of convenience and versatility.

The design of the rotating axle took into account how much mobility our client would have while on the bike. While a clamping mechanism would be important, a suitable feature of the design would be if it didn't require a lot of work to clamp onto the wheel. The Irwin clamp can provide two very useful features. First, the user would be able to clamp it on and off the bike while only using one hand. This feature allows the user to use only one hand to clamp or unclamp the wheel, while using the other hand to hold onto the handles or another part of the bike, thus increasing her stability. Second, while rotating the axis back and then turning the clamp backwards, the clamp can be clamped onto the bicycle frame. This moves the clamp out of the way when pedaling, and makes sure it stays secure on the bike. The maneuverability of the rotatable axle allowed for the feature of the clamp clamping onto the frame and out of the way of the pedals.

Future Development

To maximize the potential of the ClampMaster 2000, a few changes need to be made. The current design is mostly unable to , the product should be adjusted and remanufactured, changing materials, measurements, and more. Specifics of the changes include:

- *Alternative materials:* An aluminum build would allow the product to have improved durability, at the same level as that of the bike meaning that use of the product will not cause significant damage to the frame of the bike. However, due to the possibility of aluminum corroding over time in harsh conditions, coating the product in rubber would allow for much stronger weather resistance, as well as further prevent any scratching of the bike frame by the product. Rubber coating the design in the same color of the bike frame will also allow for a more subtle look, ensuring that the design does not attract too much attention. A possible concern regarding the aluminum build is the weight of aluminum, which may violate our requirement for a lightweight build for portability. However, this issue can be mostly avoided without sacrificing the durability of the product by partially hollowing out the design, similar to that of the inner tubing of aluminum bikes.
- *Adjusting measurements:* Particularly, the clamp for the frame of the bike needs to be adjusted, as the bike used for testing will not be the bike used by the user in the future. The width of the gap for the clamp should match the thickness of the user's bike.
- *Better storage/locking mechanism:* Due to the high possibility of bike stealing on campus, a better locking mechanism should be considered, regarding not only the bike

itself but its separate components as well. Assuming that the patient is to purchase a tricycle, it may be an easy target to those seeking bike wheels.

Conclusion

To design a method to support our client—a 17-year-old rising college freshman with cerebral palsy—mount a bike by eliminating the swivel of the handlebar during mounting, our team came up with and designed the Frame Lock. Containing three main components, the Frame Lock is a clamping mechanism attached to the frame of the bike, with a rotating lever and another clamp firmly clamping onto the front wheel of the bike. While not in use, the Frame Lock remains out of the way to promote maximum comfort and safety while biking. Featuring a small, portable, and aesthetically pleasing design along with a durable structure, the Frame Lock offers an effective solution to the given problem by fully preventing handlebar rotation while in use.

Throughout this process, our team enjoyed brainstorming unique solutions to this problem, spending time tinkering with different materials while designing our prototype, and felt great satisfaction in watching the whole project come to life. We look forward to seeing the future outlook of the Frame Lock, such as optimizing the design in future works and being able to offer the strongest solution we have for our client.

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Appendix A: Background Research Summary

Our project involves designing a method for a Shirley Ryan Ability Lab patient with Myelomalacia and spastic cerebral palsy in her legs to mount a bike. The patient's current method requires the handlebar of the bike to remain completely stationary, with no swivel; designing a product that eliminates this swivel safely and consistently suffices as a proper solution. Additionally, the bike is to be used in the future for transportation from class to class and daily activities in college, as the patient is entering college next year. This creates additional requirements for the project, including a subtle design that doesn't attract too much attention, weather resistant materials, etc.

Our team conducted background research to better understand the conditions affecting the patient, current bike modifications and components, and other miscellaneous materials. Research was conducted using major online databases such as Northwestern Library Databases like NUSearch, Google, etc. Through this process, we gathered beneficial information regarding the severity and extent of the patient's conditions, using aluminum as the primary material for our solution, and varying sources of possible inspiration for our product.

Background Research of Myelomalacia and Cerebral Palsy

Myelomalacia

Myelomalacia is a spinal cord injury causing loss of motor function or sudden jerks in the limbs [1], commonly caused by internal hemorrhaging or poor blood supply, resulting in weakness. Spinal cord damage typically results in filling of white blood cells in the damaged area, creating gray matter resulting in necrosis and loss of function, which may expand over time [2].



Figure A1: Accumulation of Gray Matter in Spinal Fluid [2]

Symptoms of myelomalacia vary highly depending on the damage done and level of necrosis within the nerves. In our case, the patient still has some level of function in her legs, able to walk with reduced kinesthetics, seemingly indicative of a lower level of damage. However, myelomalacia has no known treatment and can increase in severity over time, increasing the amount of damage caused to the body[3].

Cerebral Palsy

Cerebral palsy (CP) is a group of disorders appearing in early childhood, typically caused by lesions or damage of white matter in the developing brain, usually during pregnancy or childhood[4]. CP can also be caused by severe loss of oxygen or intracranial hemorrhaging. Symptoms of CP include ataxia, spasticity, difficulty with precise movements, and other difficulties with muscle coordination. In less common cases, CP can cause impairments in vision, hearing, and intellectual functioning [5]. CP is generally split into four types[5]:

- Spastic CP: Stiffness/awkwardness of movement, typically in one side of the body or arm, but can include legs as well, which can lead to early development of scoliosis in the spine. The most severe form of spastic CP results in full quadriplegia.
- Dyskinetic CP: Uncontrollable hyperactivity in muscles, causing twitching, writhing, etc.
- Ataxic CP: Damage to senses of balance and depth perception, causing poor coordination and involuntary movements.
- Mixed types: CP with symptoms of multiple types.

While no treatment for CP exists, therapy is commonly used to improve a patients' condition. Common methods of therapy include: physical/recreational therapy, occupational therapy, speech and language therapy, etc [5]. Orthotics are also commonly used to compensate for muscle imbalance, improving overall mobility. Our patient uses ankle foot orthotics (AFO), which primarily assist by maintaining and stabilizing alignment between the ankle and foot. AFOs typically are made from plastic and come in a few different forms: solid (SAFO), posterior leaf spring orthotics (PLSO), patellar tendon bearing (PTB-AFO), walking boot (WB), etc. Each type has a different function, but nearly every type functions to eliminate most/all ankle movement, heavily restricting kinesthetic ability within the lower leg [6]. This serves to stabilize the user's posture, but also creates further stiffness in the lower legs. This stiffness rules out a lot of options for designing an alternative method for the patient to mount the bike, and seems to point at supporting the patient's current method of mounting the bike as the best solution.

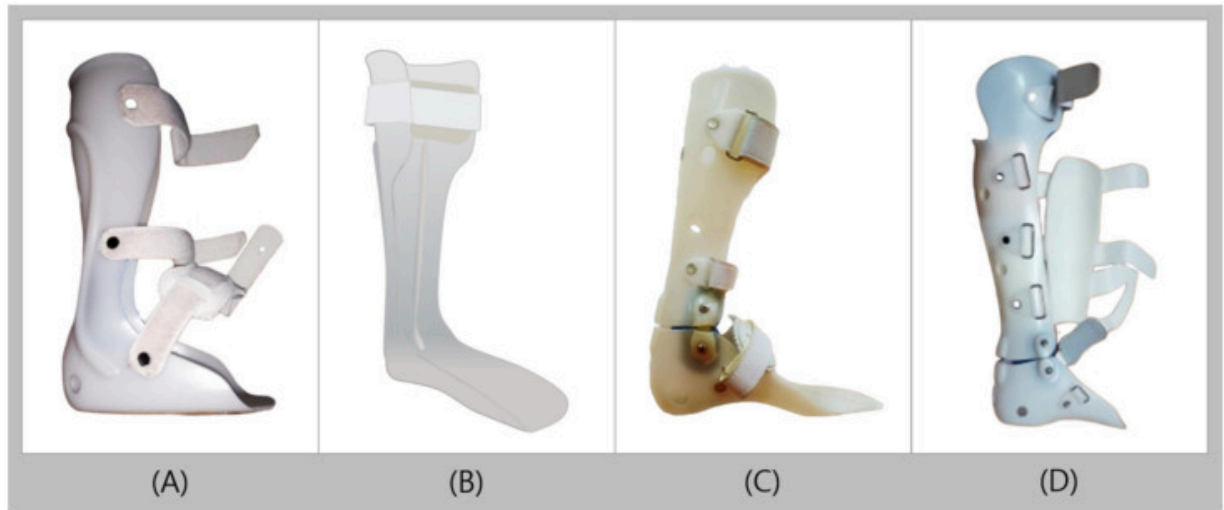


Figure A2: Types of Orthotics: A. SAFO B. PLSO C. Hinged AFO D. PTB-AFO [6]

Understanding Bikes

Understanding the different components of bikes and how they may be modified allows our team to develop a solution that is more effective, convenient, and comfortable for our client.

Bike Seat

While possibly unnecessary for our solution, examining seat modification may help us understand particular size limitations and allow us to design a more comfortable solution. Seat modification often includes changes in saddle height and width. For example, female bike seats often have wider saddles due to differences in skeletal structure, with women having wider pelvic bones [7]. Additionally, saddle height and tilt vary between male and female bikes, as females tend to display greater anterior tilt when riding a bike [7]. As a result, changing the height of the saddle leads to improved comfort and less strain on the pelvic bones.

Bike Materials

Bikes are typically quite sturdy and weather resistant, researching the materials used to create the bike may give more insight into the best materials for the solution. Most modern bikes appear to be made from some configuration of aluminum, carbon fiber, steel, or titanium, with the most common being mostly aluminum [8]. Different aluminum bikes also have varying thicknesses of tubing, including straight gauge, along with single, double, and triple butted tubings, increasing in price, respectively [8]. Assuming the bike the client decides to purchase is mostly aluminum, that limits our team's ability to modify the frame itself (i.e. drilling a hole in the frame), due to the fragility of aluminum over time. Rather, it seems optimal to create a detachable add-on, preferably aluminum to both blend in and reinforce the weatherproof potential of the product. A typical aluminum mountain bike frame appears to be able to sustain around 1200 N of horizontal

force around 90 times before showing major signs of wear [9]. This demonstrates that an aluminum build for our solution will sustain the full weight of the client, which is around 750 N, but may not be able to consistently sustain heavy modification of the frame as a solution, for example drilling a hole into the frame and fitting a metal rod through to eliminate swivel.

Adaptive Tricycle

While the client has expressed little enthusiasm for an adaptive bike due to its outstanding appearance and likelihood to garner attention, it still may be important to understand its structure and strengths. The Cerebral Palsy Foundation appears to best recommend the Rifton Adaptive Tricycle [10]. The trike appears to be extremely configurable, with adjustable handlebars, backrest, and seat height based on the user's best interest [11]. The tricycle is longer than a standard bike (ranging from 44 to 65 inches), with smaller wheels and a much lower seat [12]. Additionally, many detachables are included with the product, interestingly including a stationary stand and a pedal raise kit. The stationary stand is a set of two triangle shaped bases, attached to the rear wheels, eliminating any rotation of wheels and thus, movement of the bike [12]. The pedal raise kit allows for a taller pedal via two adjustable rods, presumably attached to the pedals itself. While designed for a tricycle, the stationary stand seems like a possible area of influence for a future solution, possibly enabling new methods for balance while mounting for more stability.

Tricycle features



Figure A3: Rifton Adaptive Tricycle [12]

**Stationary stand**

The stationary stand converts any Rifton tricycle into a stationary trainer for pedaling practice or rainy-day activity.



Figure A4: Stationary Stand for Rifton Adaptive Tricycle [12]

Miscellaneous*Trigger Clamps*

A trigger clamp is a portable, handheld version of a vice grip, featuring an easy-to-use mechanism that only requires squeezing a trigger with varying sizes and strengths, including micro (40 lbs of force), medium (100 lbs of force), large (300 lbs of force) and extra large (600 lbs of force) [13]. Due to this restrictive force, a medium or large trigger clamp may suffice as a proper method to eliminate the swivel of the handlebar if positioned properly.



Figure A5: Arrangement of Trigger Clamps [13]

Appendix B: Client Interview Summary

Introduction:

On April 5, 2024 at the Ford Motor Company Engineering Design Center in Northwestern University, our team met with Dr. Abigail Jones, the client's PT. She introduced our client as someone with cerebral palsy. She has difficulties lifting her legs around the bike and has a very particular way she has to get on the bike to ride it. What our client wants ideally is a bike that she can get on, that doesn't look obtrusive and looks like a normal beach cruiser.

Methodology:

The way she gets on the bike is that she puts her right foot on the right pedal, then gets herself on the seat, then picks up her left leg and swings it around with her left arm while her right arm holds onto the seat. Furthermore the questions we asked were:

- What difficulties do you have riding and using a bike currently?
 - Overall mobility issues
 - Could you give a demonstration
- What did you initially have in mind for the bike?
 - How do you want it to look
 - How do you want it to feel
 - Why should it feel this way
 - Is something electric important
 - How important is it for this bike to be easy to fix
 - What are things you don't want the bike to look like
 - Why
 - What school is this for
 - Is security an issue
 - Is storage going to be an issue
- What is normal
 - What does having a normal bike mean to you
 - Do you have any experience with any other things that solve this problem, if so what was your experience
 - Is a trike normal
- Specifics
 - How much do you know about bikes already
 - Is the bike just for school, or would you use it for trails/not school
 - How much is too much
 - Would something like training wheels or something akin be inappropriate
 - Would you want any changes to the bike or would you like certain extensions

- Would finding a way to make the bike handles don't twist improve getting onto the bike, or would you want a product that bypasses the problem of the handlebars all together.

Results

User Details

Dr. Jones mentioned that the client is about 5'2, with cerebral palsy and is unable to get onto the bike: it is extremely difficult for her to lift her leg up onto the bike. Her current method of mounting the bike involves putting both her feet on the right pedal while holding onto the handlebars, then physically grabbing her left leg and swinging it over. For this to occur, the handlebar of the bike cannot swivel and the bike has to be balanced, fully supporting her weight. The client also uses AFOs, which keep her feet fully straight at 90 degrees, meaning she is unable to buckle her knees. Her hip flexion is also reduced, being able to rotate no more than 90 degrees. Beyond that, there seems to be no restrictions with her upper body, but consistent overuse of her back may cause further issues down the line.

Bike Details

The primary issue is the fact that the client is unable to get onto a bike: her current method requires the bike to be balanced and for the handlebar to not swivel. The client has successfully mounted a Divvy bike, mountain bike, and upright gym bike using her current method. It's most likely that she will purchase a tricycle, eliminating the issue of balance, so eliminating the handlebar swivel would be considered a successful project. However, it is important to keep in mind that the client must be able to engage and disengage the apparatus while on the bike. Furthermore, the client mentioned her desire for a design that doesn't stand out or attract too much attention, with the design being small enough to fit in her bike basket. The bike also does not need modified pedals, as Shirley Ryan will provide unique ones. Finally, the client will likely only use the bike on campus, so there is no concern for difficult terrain concerns.

Discussion

The results changed how we viewed the problem as a whole. We narrowed it down to finding something to balance the bike while the patient is trying to get on and something that stops the front wheel from swiveling. Discussion is all about who to talk to, we plan on going to a local bike shop in Evanston, doing in person observations at Shirley Ryan, and conducting secondary research to find a solution.

Appendix C: Project Definition

Project Name: Adaptive Bike/Trike for a Patient with Cerebral Palsy

Client: Andrea

Team Members: Jieming Mei, Oba Odusote, Eytan Sayah, Barkot Simegn

Date: 04/15/24

Version: Three

Mission Statement: Create a low-profile and easy-to-use solution that allows an individual with cerebral palsy to mount a bike safely, comfortably, independently, and visually appealing.

Deliverables: A final prototype of the solution, a written report, and a final presentation to the project partner

Constraints

- \$150 budget for prototyping
- June 3rd deadline

Users & Stakeholders

- Primary: The Client who has cerebral palsy and will attend Northwestern in Fall 2024
- Secondary: Abigail Jones (Client's PT, will assist them with riding the bike independently)
- Tertiary: Shirley Ryan Ability Lab and other people who may have similar physical conditions as our client and want to ride bicycles.

User Profile

The client is a 17-year-old currently studying in high school. She is coming to Northwestern next year to study Chemistry. She has cerebral palsy and is determined to ride a bike to get to and from class. She is 74 kg, 4'9", 4'11" with her AFOs (ankle-foot orthotics) that stabilize her. She likes darker colors and wants her bike to be a single solid color. She prefers that our design does not stand out when it is on the bike.

Problem Scenario

Ideally, the client will be able to use her bike every day when going to class. She is planning on living in Bobb Hall on campus which is close to Tech where most of her classes are going to be; however, as a chemistry major, she will also have some classes on South campus. Without a bike, she would have to walk to and from classes carrying her bag and possibly other items. Her classes may be on separate parts of campus which may become a problem for her as she is not very comfortable with walking.

Solution Scenario

On a typical morning, she wakes up, gets ready for class, and walks over to Sargent Hall to get breakfast. The back doors of the buildings are so close that it doesn't make sense to get her bike. After breakfast, she gets on her bike which is parked at the racks between Bobb and Sargent Hall for easy access. This way she does not have to walk far to get to her bike even if she walks over to Sarge to get food. Using our product, she safely gets on her bike and rides to class. When she gets to the building, she locks the bike in one of the bike racks nearby and heads inside for class.

Requirements

Structure

- Must prevent the front wheel from swiveling
- Must support the client's full weight as she gets on the bike
- Must be relatively small (fits in the basket with the client's bag)
- Must be portable
- Must be visually appealing/low profile
- Must be applicable to different types of bikes
- Must be easy-to-use

Maintenance/Durability

- Must be able to support the client's body weight.
- Must be durable enough to last a long time (at least 4 years when the client is in college)
- Must be weatherproof since it will live on a bike

Application

- Must be lightweight (If it isn't permanently attached to the bike the client will have to take it off and put it in the basket with one hand after getting on the bike)

Table C1: Metrics and Specifications

Requirements	Specification	Metrics	Units	Ideal value	Allowable Value
Small/low profile/portable	Shouldn't be noticeable or get in the way	Length	in	5	8
		Width	in	3 Depends on bike, needs to wrap around frame and touch wheel	5
		Height	in	12 Depends on	15

				bike, needs to wrap around frame and touch wheel	
Easy-to-use	Easy for someone on a bike to adjust	Time taken to fully tighten and untighten	sec	10	15
Stable	Should stop wheel from swiveling	Distance wheel swivels while in use	in	0	2
Lightweight	Shouldn't significantly affect weight of bike	Weight	lbs	5 Varies based on material used	10
Durable	Needs to be weatherproof and last a long time	Time before repairs are needed	years	4	1
Sturdy	Needs to be able to sustain client's body weight	Weight sustained	kg	80	100
Aesthetic	Needs to be visually pleasing	Client rating of visual appeal	N/A, scale from 1-10	10	7.5
Adjustable	Needs to fit multiple bike models	Range of usable frame diameters	in	2	1

Appendix D: User Observation Summary

Follow the IMRaD structure to organize the summary. Keep in mind that the final report has an external audience.

Introduction

On April 12, 2024, Team 1, consisting of Barkot Simegn, Eytan Sayah, Jieming Mei and Oba Odusote, traveled to Shirley Ryan Ability Lab, where they interacted with and performed testing with the client and Dr. Abigail Jones, her physical therapist. We asked questions about the client's desired solution and examined the extent of the client's condition, discussing what she is physically able and unable to do. We also were able to record the client using her current method to mount a bike and noted possible points of concern.

Methods

The research process first consisted of us introducing ourselves to Abigail, where we learned a bit about her and how she would like her time to be at Northwestern. We asked her questions about what she would like her design to include, that way we would be able to incorporate that into our design, after the initial testing, she showed us how she got onto a bike.

With her permission we were able to record the process that she gets onto the bike, and upon reviewing it, we were able to see the struggles, the front section of the bike not staying stabilized and were therefore able to eliminate mockups that wouldn't help as much, like finding a different place for her to mount the bike, and expand on different parts of our mockup that would be able to help, keeping the wheel stationary and the different ways we would be able to achieve that.

Results

Different things we observed and saw are listed in the table below:

Table D1: User Observations and Comments

What We Saw	What Was Said About it
We observed her ankle guards closely, sketching them, photoing them, and observing how she uses them to walk, get on the bike, and live her life.	She gets taller with her ankle supports on, additionally, she uses the front handle, which is an elastic band around the front of the support keeping it in place, grabs on to it, and swings her legs around the bike either in front or behind to get on the seat.

She needs a strong foundation of support to be able to get her legs around. She can slip on occasion, especially with smaller petals, causing her to fall off. She grabs onto the front handle bar for support with her left hand. While trying to get on, she needed two people, one to stop the swiveling, and one to keep the pedal from sliding underneath her or her weight causing the bike to move forward. She brought out a mobility tricycle, that the patient had a much easier time riding, with wide pedals and a low front beam. We observed our client get onto a divvy bike with limited success, she managed to get on, but would need someone to hold the petals so she can not slip.	The petals were very small, they said that they had looked into petals that work for them, however, she did say that having some sort of cage, not just a wider petal, would help in riding the bike and petaling. She uses her right hand as mentioned earlier to swing her leg around. However, she uses her left hand as well to support herself while she is mounting the bike. These were identified as the largest struggle areas for our client. As while she is struggling to get on her bike, she needs the bike to not move, these became our areas of focus during the latter half of the observation. This tricycle we found out was about 4,000 dollars, which is well out of everybody's price range involved, however key takeaways are finding bikes with low frames and wide pedals She was able to get on, and we were reassured that this was out of the ordinary what was being demonstrated, but it was interesting to note the petals, we talked about it some more afterwards and something to keep the pedals from moving would be nice.
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She struggled to keep her feet on the petals of the divvy bike once she got on	This indicated to us that her saddle should be further back on the bike and the petals further away from her. She also expressed interest in having a lower range of motion for the petaling.
She is 4'11" tall	This leads to a bit of an issue because she needs a tall saddle to get on the bike, but she is 4' 11" so having a tall saddle leads to her not being able to reach things sometimes, including the saddle. Some sort of saddle lift system might benefit, however that would either be expensive or dangerous.
She can very easily get onto a scooter	This was talked about as a collective, about how scooter bikes might be a good solution for her, however they are unstable and expensive so more research has to be done.

Discussion

Overall takeaways, we got a lot of good recordings and measurements, including those of the client and the way she mounts a bike currently. Overall, the project is taking a new direction and there is new life to this. Sketches and initial ideas are coming out; we have started talking about manufacturing methods and brainstorming things that might work, while keeping in mind the limitations of our client and what our client wants. It is possible for the client to reach down relatively far, widening the extent of a possible solution. It seems that as long as the solution remains on the bike, and effectively eliminates the swivel of the front handlebar, it should remain as an effective solution. More thorough notes have been taken by all members on the trip and all that's left to do now is more research, more designing, and starting to think about initial mockups and how to move on.

Appendix E: User Feedback and Testing Summary

Introduction

On April 30th 2024, our team went into Shirley Ryan Ability Lab to talk with our client and her therapist about our initial ideas and where we can iterate from where we were at the time. We did this using mockups we had built the week prior, tested it with our client and then proceeded from there.

Methodology

Each mockup was introduced one by one, describing to the client the inner workings of the solution and letting her carefully prod and observe up close. Following the initial introduction, we tested each mockup on a small bike at Shirley Ryan, carefully describing how each design would work and finally demonstrating each mockup in action, attaching each to the bike and physically testing it. However, due to the fragility of the mockups, we did not have the client herself fully test each mockup, fearful for possible injury. Following the basic testing stages, we then broadly asked the client and her PT about their thoughts on each design, then asked for further specifics, such as how well the client feels that the design fits her needs for aesthetics, ease of use, stability, etc. This ultimately led to a discussion about what could be improved to each design and what new ideas should be introduced to each design. Testing each mockup took roughly 20 minutes of time.

Results

The Drop Stop

This idea came from a less refined version of the prompt we were given, and a more ambitious desire to try and solve as many problems as possible. It works by having a heavy wooden block attached to the central hub of a bike's front wheel. When testing, we found that it didn't work as expected in the slightest. Rather, it would slip constantly, would require an external device to raise and lower it, and is cumbersome and makes the bike difficult to ride. While there were many failures from this design. Some key insight came from it. Focusing on locking the front wheel is the easiest and most efficient way to help our client get on and off the bike. Thus from this failure came new focus. Find a way to Stop the rotation of the front wheel

Seat Grip

Tangentially related to all our other ideas. We brought forward a novel idea that wasn't like anything else we had come up with or were thinking of building to see if our client would like it. The result. A mixed bag. On the one hand. Having a handle on the bottom of the seat was helpful to our client and made the job of getting on and off the bike mildly easier. But what we discovered was that the aid it provided was marginal and dependent on the type of bike. The type of bike we had available was one in which it would help, but explanation from our client's

therapist showed that most of the types of bike they were considering were not going to be made easier to mount with the seat grip. So we scrapped it.

Frame Lock

This was the most promising idea by far. Attach some sort of arm to the side of the bike's frame and use it to keep the bike in place. It worked but not ideally. It required an additional bolt threaded through the bike wheel. Making it impractical for any mobile application. Furthermore the grip would slip off the frame of the bike from time to time but there was promise. This was the idea our client and their therapist enjoyed the most. Taking our mockups and running away with ideas as we frantically jotted down notes. The success came from how small it was and the novel way that we stopped the bike's front wheel. Not by stopping the entire bike or groping the handle bars, but going right for the wheel itself.



Figure E1: Presentation of the Frame Lock and Suggestions upon Improvement

Discussion

After talking with our client, her therapist, and consulting our professors. We came to the conclusion that the best way to help our client was to make it as easy as possible for her to stop the front wheel from spinning. From that we went back to the shop to prototype, started playing around with some vice grips, eventually came upon some clamps, and that inspired our current design.

Appendix F: Instructions for Use

The following instructions will allow a user to effectively and efficiently attach, engage, disengage, and remove the ClampMaster 2000 for easy use, taking only 3 minutes to complete

Step 1: Gather all given materials: the ClampMaster itself along with the 4 bolts and washers. put the clamp around the desired part of the frame and then thread each bolt through each hole as shown in Figure F1. Then, tighten each nut onto its corresponding bolt, ensuring that there is no wiggling or instability while the ClampMaster is on the bike. When completed, the apparatus should look like Figure F2.

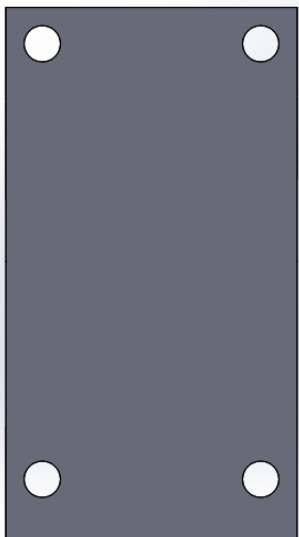


Figure F1: Bolt Locations



Figure F2: Completed Initial Attachment

Step 2: To engage the clamp, rotate the clamp such that it is positioned right besides the front wheel, and simply squeeze the trigger on the clamp until it firmly attaches to the front wheel.



Figure F3: Fully Engaged ClampMaster 2000

Step 3: Mount the bike.

Step 4: To disengage the clamp, simply pull on the clip on the clamp itself, then pull the clamp back until it no longer clamps onto the wheel. The ClampMaster 2000 should automatically remain out of the way of any standard biking movements.



Figure F4: ClampMaster 2000 Fully Disengaged

Step 5: To remove the apparatus, simply remove the nuts from the frame connector and pull out the bolts.

Appendix G: Design Review Summary

Introduction

On 5/9, 2024, Team 1 presented our current idea to solve the bike mounting problem to DTC Section 22 for design review. Our team presented the problem at hand, introduced our main idea and mockup, and discussed our main concerns with the project. We asked the section for advice in regards to what material should be used for the design, how the design should be stored when not in use, which clamping mechanism would be most effective, and other miscellaneous advice.

Table G1: Design Review Summary

Reviewers Liked	Reviewers Disliked	Features to be Added	Features to be Modified	Additional Comments
<u>Safety</u> General design satisfactory Durability under standard conditions <u>Versatility</u> Usable for multiple bikes	<u>Potential Hazards</u> Puncture possibility Concerns about ability for patient to reach Durability in wet/harsh conditions	Strings to restrain movement when not in use(?) Any method to restrain while not in use	Possibility of designing a wheel sleeve Clamping system	<u>Presentation</u> Better explanation of problem in future, state key areas of design <u>Materials</u> Rubber clamp(?) <u>Aesthetics</u> Keep in mind, think about size concerns

Future Plans

Our team will continue to do research and decide on the ideal material for our design, and reconsider certain aspects of the design, incorporating discussions about particular hazards that may arise with the design. We will try to incorporate the idea of the claw clamping onto the frame of the bike when not in use, and modify our current clamping mechanism to a smaller one.

Appendix H: Performance Testing

After constructing the prototype, we tested for effectiveness and any possible issues during the use of our device. From May 20th-June 3rd, we tested the device's durability, ease of use, and strength with a few random volunteers. To test it, we would hold the bike from the back and let our subject try to get on the bike with all their weight at once, bike around, then get off in the same manner they got on.

Table H1: Prototype Testing and Feedback

Testers	Results
1 165lbs	Held up there weight, would sometimes get in the way of biking due to having a larger frame.
2 125lbs	Worked without any failure, got it first try without an explanation, held up there weight just fine.
3 130lbs	Confused for a long time on what it was. Said "if you guys just hold up the bike why would it matter if the front wheel is locked or not"
4 145lbs	Really gave it his all to break it, had us worried, but the design held. Even out of just PLA

Feedback

Unimanual use was a big selling point to those that tried it. There was a bit of a learning curve, but that usually came as a result of a novel concept. Once people were shown once, they understood immediately. One pressing issue is that for people with large bodies and longer frames, feet and shins would sometimes come near to the device when just biking, this can be remedied with either a smaller newer model or experimentation with placement on different types of bikes to optimize a design for a broader spectrum of bikes and people.

Appendix I: Instructions for Construction

The following instructions will allow a user to create a working version of the ClampMaster 2000, using aluminum and a given clamp.

Step 1 : Download the following DXF files from [this link](#):

- Top half Wheel Clamp x1
- Connector x1
- Bottom half Wheel Clamp x1
- Bike Clamp x1

Step 2: Gather 5 blocks of 3x7x7 in aluminum, then for each block, then download each DXF file from the Step 1, then water jet a copy of each of the DXF files, cutting out the 5 components of the total block.



Figure I1: Major Components of the ClampMaster 2000

Step 3: To allow the bike clamp to separate as illustrated in Figure I1, we need to cut 4 holes in the corners using a dowel to punch the holes in, then bore the holes using a lathe. The holes need to be half an inch in diameter to allow stock machine screws to fit in the holes and allow the Bike clamp to be closed together.

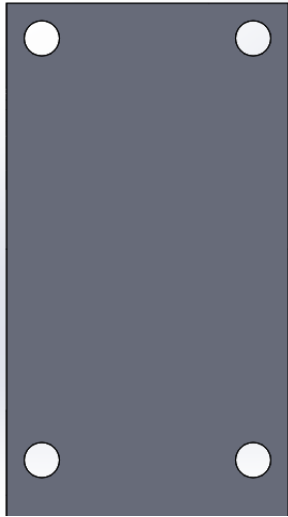


Figure I2: Connector Holes

Step 4: The connector will have a thin overhang, cut a 1 inch hole in the center of each of these overhangs, using the same method as Step 3.

Step 5: Cut the connector in half using a water jet to get the two halves separated (Figure I1), allowing for the connector to sandwich onto the bike frame.

Step 6: Purchase a clamp from [this link](#), then unscrew and take apart the clamp, leaving one side with the bar, threading the rotating axle through this bar, then reattaching the rest of the clamp.

Step 7: Position the holes in the top of the rotating axle in line with the holes of the overhangs, then firmly bolt a screw through each side.



Figure I3: Completed ClampMaster 2000

Appendix J: Bill of Materials

Table : Bill of Materials

	Item	Distributor	Price
1	PLA 3D Printing Material	Northwestern	\$0
2	Clamp	Amazon	\$10
3	Aluminum 7x7x3 blocks	McMaster Carr	\$700